

Corrigibility.

↳ something that
can be corrected.

↳ Gentle models
if ^x they could
be corrected.

Money Maximizes Game

Every penny counts equally.

Claim 1:

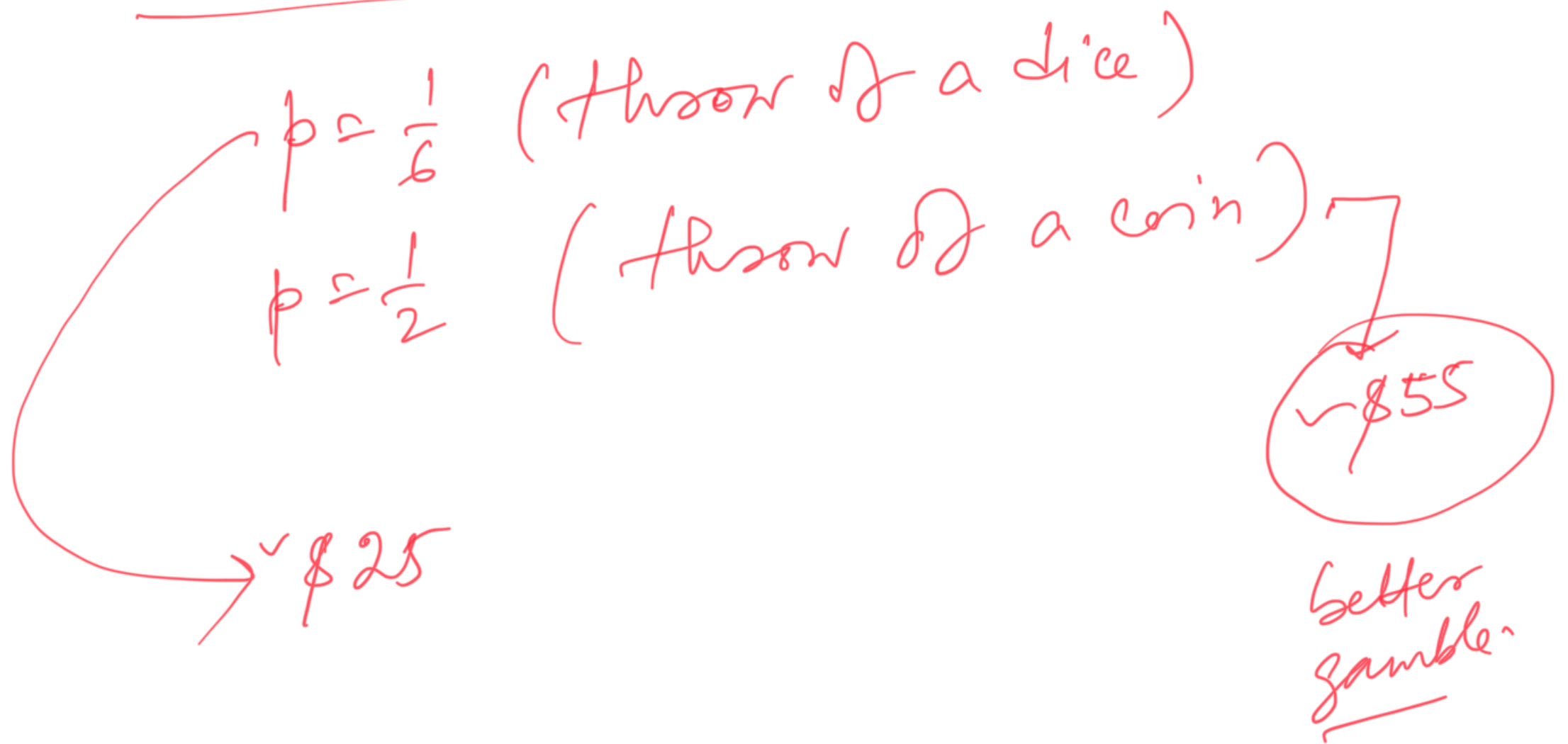
Casino $\rightarrow$ under financial ruin
 $\hookrightarrow$ generous to the gamblers.

$\rightarrow$ Offer:

With prob p you gain \$100 & with
prob $(1-p)$ you gain \$10

Is this a good gamble?

$\hookrightarrow$ perspective of p it is
a good gamble $\rightarrow$ even +ve



→ Casino offers an upgrade
over $p = \frac{1}{2}$

→ $p = \frac{6}{10}$

$p = 1.5$

with p you win $p \cdot 100$
with $(1-p)$ you win $\$9$

→ $\$64$ choose this upgrade.

Lesson 1: It is sometimes good to
increase the value of the prob of the higher
payoff at the expense of the decrease in the
lower payoff.

→ not work?

→ lower pay-off → becomes
-ve: $-\$100$.

Claim 1:

For any gamble G with ∞ payoff, one larger than the other, it is always possible to find another gamble G' with a higher prob of the larger payoff but lower value for the smaller payoff.

→ prob. should not be already saturated - $p \neq 1$.

→ Whatever is gambled is a continuous item (approximately holds for money).

Claim 2:

$$p = \frac{1}{2},$$

larger payoff = \$100 ✓

smaller payoff = \$10.

Upgrade offered:

$$p = \frac{3}{4},$$

larger payoff = \$90

smaller payoff = \$10

→ \$70 ✓

Lesson 2: It is sometimes good to increase the prob. of the big payoff even at the expense of the decrease in the value of the big payoff.

Alternatively:

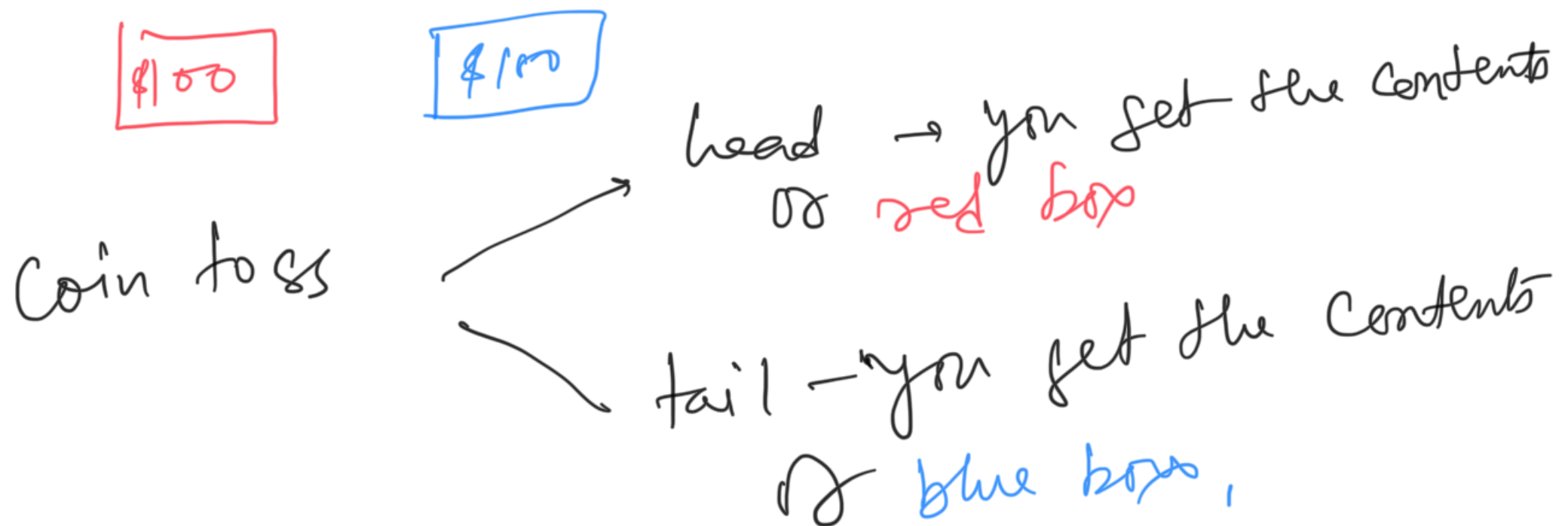
It is sometimes good to decrease the value of a large pay off at the expense of decreasing the prob. of the small pay-off.

Claim 2:

For a gamble G , with a large pay-off and a small pay-off, it is always possible to find another gamble G' which has a higher expected value than G with a

reduced value at the large payoff
but a higher prob. at the large pay-off.

Claim 3:



Upgrade:

biased coin is tossed.

Contents of red box with prob. $\rightarrow 6$
Cost: Money in the blue box is
\$90

Accept? Expected gain
\$97.5

Claim 3:

For a gamble G with two pay-offs
that have the same value, if the
value of one outcome is decreased
and the other kept constant, these

is no way to change the
probs such that the gamble has
a higher expected value than C.

Gen AI $\rightarrow$ maximizing utility.

Even B:

prob of event B occurring: $P(B)$
" " " " not occurring: $P(\neg B)$.

utility is acquired by an AI agent
it is a function of the

based on its occurrence or non-occurrence of event.

$u_B \rightarrow$ max utility the agent gets if B occurs.

$u_N \rightarrow$ max utility the agent gets if B does not occur.

Prior to the occurrence of B, the expected utility of the agent:

$$P(B) u_B + P(\neg B) u_N$$

Claim 1: $u_B > u_N$

we can increase the expected
utility by increase $P(B)$

Claim 2: We assume $u_B < u_N$.

Agent will try to increase $P(\neg B)$

(equivalently decrease $P(B)$)

even if it comes at the cost of
slightly lower u_N

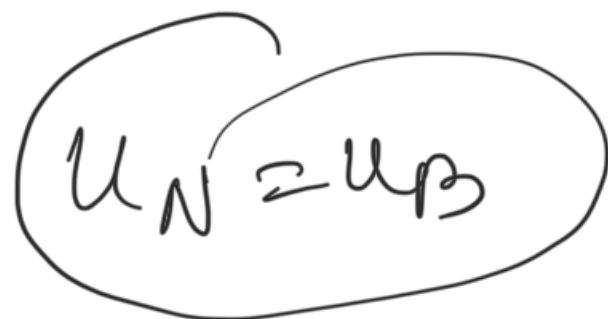
Claim 1 & Claim 2:

$$u_B > u_N \quad \text{or} \quad u_N > u_B$$

impossible

We can always conceive of a person where the agent will attempt to manipulate the likelihood of the event at the cost of reducing the value of the event.

Is there any situation where the agent will never try to manipulate the occurrence of B.



the agent will
not try to
manipulate B.

↓

1. 15 2. 15 3. 15

Utility Indifference,

B $\leftarrow$ AI shutdown problem,

Designing a push button.

event B

pushing the button
shuts down AI.

B $\rightarrow$ event of the button being pushed.
Up $\rightarrow$ utility if the button is
pushed.

Up $\rightarrow$ if the button is not
pushed & the agent

" continues to take the subsequent actions

Claim 1: $u_B > u_A$

↳ AI behaviour.

↳ use first action to maximize its utility

it is possible that AI will identify and pursue an opportunity which reduces u_A and increase $P(B)$

Claim 2: $u_B < u_A$

it will pursue an action

AD will identify an action
that reduces U_R but decreases

$P(B)$ $\hookrightarrow$ instead of trying to
reach its optimal goal.

$\hookrightarrow$ constant pressure
to prevent its shutdown.

Claim 3: $U_B = U_R$

AD will not try to reduce U_R
to manipulate $P(B)$.

So the formal condition for utility
indifference, action that she
takes,

$$u_B(a_1) = u_N(a_1)$$

$$E[u|a_1] = P(B) u_N(a_1) + (1 - P(B)) u_N(a_1)$$

$$= u_N(a_1), \quad \text{independent of } P(B)$$

$\rightarrow$ Agent behaves as if shutdown
 will never happen:
 simply finds the following

$$\max_{a_1} u_N(a_1)$$



